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CAH refers to a group of autosomal recessive diseases, each associated with defective production of one or more steroid hormones by the adrenal glands, two crescent-shaped glands located above each kidney in the abdominal cavity. It has been reported that unequal crossing over involving chromosome 6 can result in CAH. More than 90% of CAH diagnoses are associated with a deficiency of 21-hydroxylase, an enzyme encoded by the *CYP21A2* gene on chromosome 6.

Normally, 21-hydroxylase catalyzes chemical reactions involved in the synthesis of aldosterone and cortisol, the principal mineralocorticoid and glucocorticoid hormones, respectively. As a result, patients with 21-hydroxylase deficiency exhibit diminished serum levels of aldosterone and cortisol. However, synthesis and secretion of the sex hormone testosterone is excessive. Among the normal actions of glucocorticoids are metabolic and anti-inflammatory effects, and glucocorticoid deficiency disrupts the negative feedback loop linking the brain to the adrenal gland. Glucocorticoid deficiency can lead to hyperplasia (increased cell proliferation) of adrenal tissue.

The patient's 21-hydroxylase deficiency is treated with regular administration of synthetic steroid hormones. Her condition improves after receiving these medications.

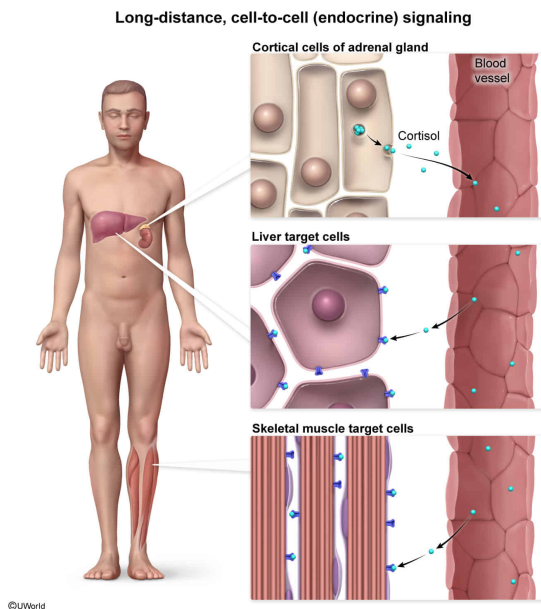
The effects of glucocorticoid deficiency are known to include impairments in superficial wound healing, memory formation, and other physiologic phenomena. This deficiency most likely leads to these wide-ranging effects because:

- ☐ A. unregulated gluconeogenesis leads to hyperglycemia that damages many of the body's tissues. [13%]
- ✓ ☒ B. receptors for glucocorticoid hormones are expressed in nearly all cell types in the body. [62%]
- ☐ C. glucocorticoids increase inflammation in most tissues, leading to diverse effects. [11%]
- ☐ D. hyperplasia of adrenal tissue in CAH causes damage to several nearby organs. [13%]

Correct

60% answered correctly

Explanation:



Hormones are **chemical messengers** that influence physiologic function through **endocrine signaling**, a form of long-distance, cell-to-cell communication. In **endocrine signaling**, hormones are secreted into the bloodstream by specialized cells and transported throughout the body via the circulatory system. Once the hormone reaches its target tissue, it alters cellular function in one of the following ways:

- **Water-soluble (hydrophilic)** hormones generally bind a cell surface receptor and initiate a second messenger cascade, leading to changes in protein activity (eg, enzymes, membrane channels).
- **Water-insoluble (hydrophobic)** hormones generally diffuse across the lipid bilayer of the plasma membrane and bind to a cytosolic receptor to ultimately regulate gene expression in the nucleus.

Hormones affect the function of diverse and distant tissues in the body. This occurs because hormone receptors can be expressed in a variety of cell types located throughout the body, and any cell that expresses the correct receptor is capable of responding to the receptor's ligand (hormone). For example, **glucocorticoids** released into the bloodstream from the **adrenal cortex** influence the energy metabolism of skeletal muscle cells, liver cells, and fat cells—which, like all other nucleated cells, express **glucocorticoid receptors**.

However, in glucocorticoid deficiency, an insufficient amount of glucocorticoid hormones in the blood leads to diminished binding of these hormones to their receptors. This condition would compromise the normal function of the various cells that express glucocorticoid receptors.

(Choice A) Glucocorticoid signaling increases serum glucose levels by promoting **gluconeogenesis**; therefore, glucocorticoid deficiency would contribute to hypoglycemia, *not* hyperglycemia.

(Choice C) As stated in the passage, glucocorticoids (eg, cortisol) generally have *anti-inflammatory* effects.

(Choice D) Tissue damage to nearby organs caused by adrenal gland enlargement would have *no* direct effect on activities such as memory formation and the healing of superficial wounds because these functions occur *outside* the abdominal cavity.

Educational objective:

Hormones are a diverse group of molecules that participate in endocrine signaling, a form of long-distance, cell-to-cell communication. Any cell type that expresses a hormone receptor will respond to serum levels of the hormone.

Time spent:0 sec

Subject:

Biology

QID:401140

System:

Endocrine and Nervous Systems

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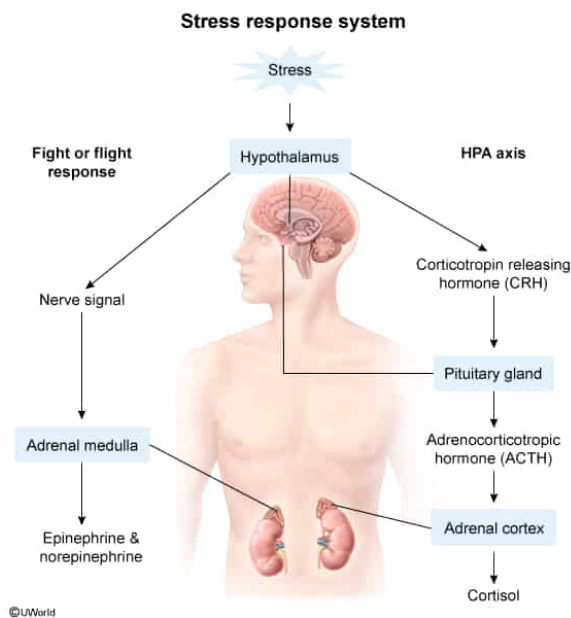
The neuroendocrine signaling cascade that induces adrenal hyperplasia in patients with 21-hydroxylase deficiency involves coordinated activity among which organs?

- ☐ A. Hypothalamus, parathyroid glands, adrenal glands [5%]
- ✓ ☒ B. Hypothalamus, pituitary gland, adrenal glands [88%]
- ☐ C. Liver, pituitary gland, adrenal glands [2%]
- ☐ D. Pituitary gland, parathyroid glands, adrenal glands [3%]

Correct

87% answered correctly

Explanation:



The **hypothalamus** is a brain structure located inferior to the thalamus that regulates the synthesis and secretion of multiple classes of hormones. It serves as the interface between the nervous and endocrine systems by processing both cortical inputs from the brain and sensing the plasma concentration of numerous hormones. In response to these inputs, the hypothalamus controls body-wide endocrine function by influencing activity within the **pituitary gland**, an endocrine organ located inferior to the hypothalamus that is sectioned into a posterior lobe and an anterior lobe.

The **posterior lobe of the pituitary gland** contains the axonal processes of hypothalamic neurons. The hormones **antidiuretic hormone (ADH)** and **oxytocin** are synthesized by these specialized neurons and stored within their axon terminals, which form junctions with systemic blood vessels. As a result, the firing of hypothalamic neurons results in the release of ADH or oxytocin into the systemic circulation.

The hypothalamus also controls the **anterior lobe of the pituitary gland** by releasing signaling molecules into the hypophyseal portal system, a localized network of blood vessels that bridges the hypothalamus and anterior pituitary. These signaling molecules promote or inhibit the release of specific hormones into the systemic circulation by the anterior pituitary gland. For example, **corticotrophin-releasing hormone (CRH)** released by the *hypothalamus* stimulates the *anterior pituitary* to release **adrenocorticotrophic hormone (ACTH)**. ACTH then acts on the *adrenal glands* to promote synthesis and secretion of glucocorticoids, completing the neuroendocrine pathway that regulates the serum level of **cortisol** and other glucocorticoids.

(Choices A and D) The **parathyroid glands** secrete **parathyroid hormone (PTH)**, which functions to regulate serum Ca^{2+} levels. The parathyroid gland does *not* directly regulate the synthesis and secretion of glucocorticoids.

(Choice C) Although the liver is responsible for **numerous functions** (eg, carbohydrate and fat metabolism), it is *not* involved in neuroendocrine control of the adrenal glands.

Educational objective:

The hypothalamus is a brain structure that regulates numerous endocrine functions through its association with the pituitary gland. Together, the hypothalamus and pituitary gland regulate the synthesis and secretion of hormones that influence metabolism, reproduction, and other important functions.

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Biology

QID:401141

System:
Endocrine and Nervous Systems

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Patients with CAH exhibit hypoglycemia due to numerous endocrine abnormalities. In healthy individuals, hypoglycemia triggers an increase in the serum level of:

- I. insulin.
- II. epinephrine.
- III. glucagon.

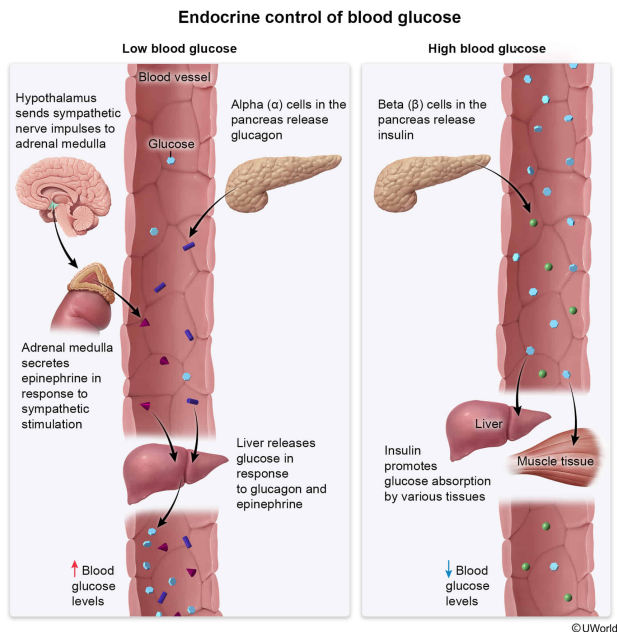
- ☐ A. I only [7%]
- ☐ B. II only [2%]

- ☐ C. I and III only [6%]
- ✓ ☒ D. II and III only [83%]

Correct

82% answered correctly

Explanation:



Homeostasis is the process by which an organism maintains a relatively stable internal environment in response to external or internal changes. For example, blood pressure, blood pH, and **body temperature** are all maintained within specific ranges that are optimal for normal bodily function. Failure to achieve homeostasis of one or more parameters can lead to disease and, potentially, death.

The level of glucose in the blood is also subject to homeostatic control primarily through the actions of hormones. When blood glucose levels are low (eg, during *hypoglycemia*), the hormone glucagon is secreted from the **alpha cells** of the **pancreas**, and the **autonomic nervous system** promotes the release of **epinephrine** and **norepinephrine** by the **adrenal medulla**. Glucagon is a peptide hormone that promotes **gluconeogenesis** (the synthesis of glucose from other molecules) and **glycogenolysis** (the breakdown of glycogen into glucose). **Epinephrine** also promotes glycogenolysis. Therefore, the overall effect of glucagon, epinephrine, and norepinephrine is to promote the synthesis of glucose and its release into the systemic circulation, thereby correcting the low blood glucose level.

According to the passage, symptoms of CAH include hypoglycemia (low blood glucose). In healthy individuals, hypoglycemia can be expected to increase the release of **glucagon**, **epinephrine**, and norepinephrine because these hormones

act to restore the glucose concentration to the normal range (**Numbers II and III**).

(Number I) In response to high blood glucose concentration, the peptide hormone insulin is secreted from the beta cells of the pancreas. Insulin promotes the uptake of glucose by fat, liver, and skeletal muscle cells. Insulin also promotes **glycogenesis**, the synthesis of the glucose storage molecule glycogen. Overall, insulin decreases blood glucose concentration by promoting the cellular uptake and storage of glucose. Because of this, hypoglycemia in healthy individuals does *not* trigger an increase in the serum level of insulin.

Educational objective:

Glucose is a key energy source for many tissues in the human body. The blood glucose concentration is tightly regulated through the actions of the hormones insulin, glucagon, epinephrine, norepinephrine, and the glucocorticoid cortisol.

Time spent:0 sec

Subject:
Biology

QID:401142

System:
Endocrine and Nervous Systems

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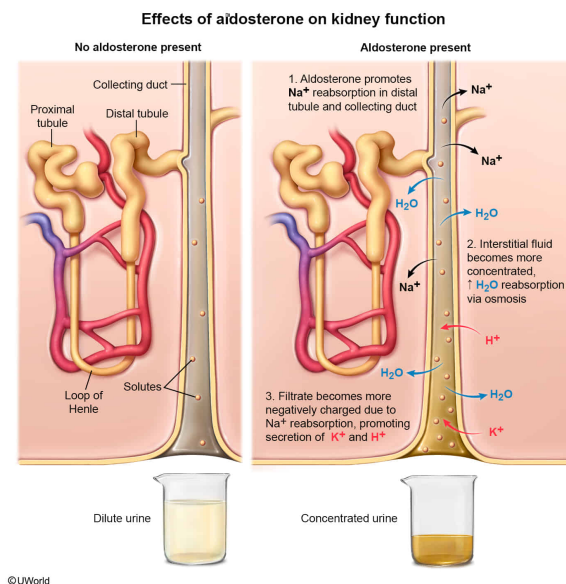
What changes in blood volume and serum sodium content would be expected as a result of using synthetic hormones to replenish aldosterone in patients with CAH?

- ✓ ☒ A. Increased blood volume and increased serum sodium content [62%]
☐ B. Increased blood volume and decreased serum sodium content [23%]
☐ C. Decreased blood volume and increased serum sodium content [8%]
☐ D. Decreased blood volume and decreased serum sodium content [5%]

Correct

62% answered correctly

Explanation:



Located in the posterior abdomen on the left and right sides of the body, the **kidneys** are two bean-shaped organs that participate in excretory, metabolic, and **homeostatic** functions. The homeostatic functions of the kidneys occur primarily through the filtration of blood and the selective **secretion or reabsorption** of filtered contents.

Filtration of blood by the kidneys yields a fluid (called the filtrate) that undergoes significant changes in volume and composition prior to excretion as urine. Nutrients such as glucose and amino acids are reabsorbed and toxins are secreted. The majority of filtered sodium (Na^+) and water (H_2O) is reabsorbed. However, the reabsorption of sodium and water is subject to some degree of adjustment based on physiological conditions.

For example, the hormone aldosterone, which is secreted by the adrenal cortex in response to high serum potassium (K^+) and low blood pressure (via the **renin-angiotensin system**), acts on the kidney to promote **sodium reabsorption** and **potassium secretion**. However, more sodium is reabsorbed than potassium is secreted. Consequently, an ion gradient is generated between the filtrate and the fluid in the adjacent interstitial space, causing **water to be reabsorbed** into the interstitial fluid (and eventually the bloodstream) via **osmosis**. As a result, **blood volume** and therefore **blood pressure increase**.

According to the passage, patients with CAH exhibit deficient synthesis and secretion of aldosterone. After administering synthetic aldosterone, increased sodium and water reabsorption by the kidneys will result in increased blood volume and increased serum sodium content.

(Choices B and D) Aldosterone promotes the reabsorption of sodium by the kidneys, leading to *increased* serum sodium content.

(Choices C and D) Aldosterone promotes the reabsorption of water by the kidneys (a side-effect of the reabsorption of sodium), leading to *increased* blood volume.

Educational objective:

Aldosterone is secreted by the adrenal cortex in response to elevated plasma potassium

and decreased blood pressure. Aldosterone acts on the kidneys to increase the reabsorption of sodium, which in turn leads to increased reabsorption of water and, ultimately, increased blood volume and blood pressure.

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Biology

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System:
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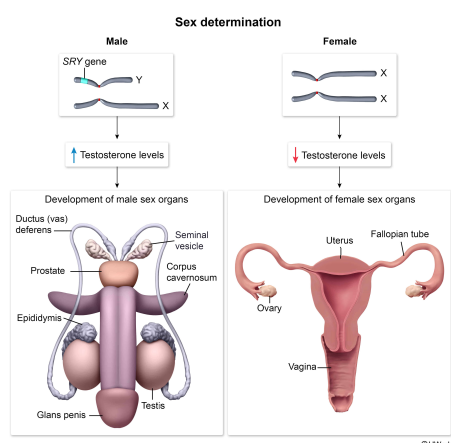
A 10-day-old male who was not feeding well was diagnosed with 21-hydroxylase deficiency despite having normal physical examination results apart from signs of dehydration. Which of the following best explains why physical examination results can be largely normal in males with this condition?

- ☐ A. The testes compensate for abnormal adrenal function by synthesizing glucocorticoids. [26%]
- ☐ B. The Y chromosome contains extra copies of genes mutated in CAH. [5%]
- ☒ C. High testosterone levels lead to abnormal sexual development earlier in females than in males. [64%]
- ☐ D. Only female sexual differentiation is dependent on serum glucocorticoids. [4%]

Correct

64% answered correctly

Explanation:



The hormones **testosterone**, estrogen, and progesterone are referred to as **sex hormones** because of their role in sexual development and reproduction. Although testosterone can be detected in the serum of females, it is considered a male sex hormone because of its role in the development of male traits and behaviors.

Testosterone production in males is related to the presence of the sex-determining region Y (SRY) gene on the Y chromosome, a sex chromosome that exists alongside one X chromosome in male cells. The SRY gene controls male **sex determination** and promotes development of the testes, the male sex organs where testosterone is produced. Testosterone within the fetal bloodstream then promotes the development of other male sex organs. Because female cells have two X chromosomes and no Y chromosome, female fetuses are exposed to less testosterone, resulting in development of female sex organs.

According to the passage, 21-hydroxylase deficiency is associated with defective glucocorticoid synthesis and excessive testosterone synthesis. In females, exposure to excess testosterone before birth would lead to abnormal development of the female external genitalia that can be detected upon physical examination at birth. Because **testosterone** is a **normal contributor** to **male sexual development**, **excess** testosterone **would not alter** development of the external sex organs **as dramatically in males**, and physical examinations during early infancy might yield **no abnormal results**.

(Choice A) Although the testes and adrenal glands both secrete **steroid hormones** (ie, hormones derived from cholesterol), the testes cannot synthesize glucocorticoids. As a result, testicular endocrine activity *cannot* correct for the glucocorticoid deficiency seen in 21-hydroxylase deficiency.

(Choice B) The Y chromosome is a sex chromosome that contains very few genes, most of which are related to male development. According to the passage, the gene for 21-hydroxylase is located on chromosome 6, an autosome. Therefore, the Y chromosome would *not* contain extra copies of autosomal genes such as those mutated in individuals with 21-hydroxylase deficiency.

(Choice D) Glucocorticoids (eg, cortisol) do *not* play a direct role in the sexual development of males or females, and therefore cannot explain the genital abnormalities seen in patients with 21-hydroxylase deficiency.

Educational objective:

Testosterone is a sex hormone that facilitates the development of male sex organs and secondary sexual characteristics.

Time spent:0 sec

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Biology

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Endocrine and Nervous Systems

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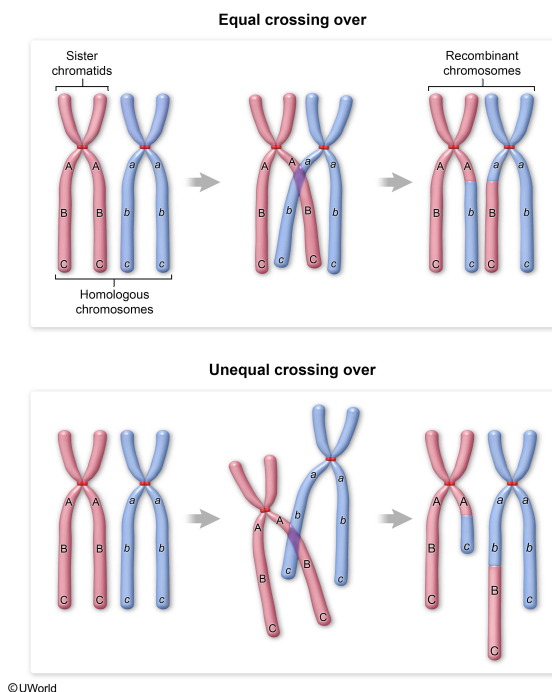
Given that the fusion of male and female gametes results in fertilization, which statement best explains how unequal crossing over during gametogenesis led to 21-hydroxylase deficiency in the fertilized embryo? One of the parental gametes:

- ☐ A. contained an extra copy of chromosome 6. [9%]
- ☐ B. contained no copies of chromosome 6. [16%]
- ☐ C. experienced an event in which the *CYP21A2* gene was duplicated. [11%]
- ✓ ☒ D. underwent an event in which the *CYP21A2* gene was deleted. [63%]

Correct

63% answered correctly

Explanation:



In eukaryotes, reproductive cells (gametes) are produced via **meiosis**, during which a parent cell ($2n$) divides to produce four genetically distinct daughter cells containing half the original number of chromosomes (n). **Crossing over**, which occurs during prophase I of meiosis, is the process by which **homologous chromosomes** swap genetic material through the breakage and exchange of corresponding DNA segments between non-sister chromatids. As a result, crossing over events give rise to **daughter cells** with chromosomes containing **allele combinations** that **differ from** those in the **parent cell**, leading to genetic recombination and **increased genetic diversity**.

When homologous chromosomes fail to align properly, **unequal crossing over** occurs. Unequal crossing over is the exchange of unequal segments of DNA between non-sister chromatids, yielding one chromosome that contains extra copies of one or more genes (ie, gene duplication) and another chromosome that lacks those genes entirely (ie, gene deletion).

According to the passage, 21-hydroxylase deficiency occurs due to insufficient activity of 21-hydroxylase, the enzyme encoded by the *CYP21A2* gene on chromosome 6. If unequal crossing over involving chromosome 6 occurs, it may result in a daughter cell that contains a chromosome lacking *CYP21A2*, which would contribute to the development of 21-hydroxylase deficiency in offspring.

(Choices A and B) **Nondisjunction** is defined as the failure of homologous chromosomes to separate in anaphase of meiosis I or the failure of sister chromatids to separate in anaphase of meiosis II. Consequently, nondisjunction produces gametes that either lack a copy of a chromosome or contain an extra copy of a chromosome. *Unequal crossing over* of genetic material between non-sister chromatids gives rise to gametes that lack or contain an extra copy of *individual genes*, not entire chromosomes.

(Choice C) 21-Hydroxylase deficiency occurs due to *insufficient* 21-hydroxylase activity, which can be caused by reduced *CYP21A2* expression or by loss of function of the 21-hydroxylase enzyme itself. Although it can result from unequal crossing over, duplication

of *CYP21A2* would cause *enhanced* 21-hydroxylase activity (*not* deficiency) due to greater expression of the gene.

Educational objective:

Crossing over is a form of genetic recombination involving the exchange of genetic material between non-sister chromatids of homologous chromosomes. Unequal crossing over can result in offspring that lack important genes.

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